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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCSDV100 -

FUNDAMENTALS OF ACOUSTICS (2024 ADMISSION ONWARD) ANSWER KEY

Part A

Very short answer Questions Answer any **10** Questions Each Question carries **2** marks

1. What is the amplitude of a sound wave ?
   Distance from the rest Position to either the crest or the trough.

2. How does sound travel differently in air compared to water?
   Sound travel more faster in water than air because the density of water is higher than the air

3. How does diffraction help us hear sounds from different directions?
   Diffraction means the bending of sound wave when it encounter an obstacle ,so sound waves can bend over the obstacle and reach our ears from distant source

4. Why are critical bands important in sound perception?
   Critical bands are important for the hearing system to discriminate between individual frequency components of an input sound.

5. What is the speed of sound in Air in relation with temperature ?
   Speed of sound increases when the temperature increases due to the change in the kinetic energy of the molecules

6. What will happen if two sound waves are In' In phase' ?
   When two sound wave are in phase they will form a new wave with double the amplitude

7. Give an example of a complex sound and briefly describe its characteristics ?
   most of the musical instruments produce complex sound waves ,Pitch/timbre/ intensity are the main factors in a complex wave

8. Describe the relationship between density and elasticity.
   Density inversely proportional to Elasticity

9. What are the crest and trough of a transverse wave?
   The topmost point of the wave medium or greatest positive distance rom the rest position/illustration

10. What is sound absorption?
    This means that vibrational energy is transferred from the sound wave to the object that has been hit.

11. What happens when two sound waves meet and interfere constructively?
    when to sound waves constructively interfere a new wave with double amplitude will form

12. What is frequency domain masking ?
    The existence of one tone effect the perception of an identical tone with less intensity

Part B

Short Answer Questions

Answer any **6** Questions Each Question carries **5** marks

13. Explain frequency range of human hearing ?
    20hHz to 20,000Hz,Sensitivity also varies with frequency, equal loudness contour,The human auditory system is most sensitive to frequencies in mid range

14. Compare the effects of too much and too little sound reflection in an auditorium.
    Delayed reflections cause reverb, time delay of reflections determine whether it heard as reverb, echo, it effects the intelligibility of sound

15. What is a wave Equation ?

- Equation in terms of velocity: $v = f\lambda$, where v is velocity, f is frequency, and λ is wavelength.

- Another Form of the wave equation:

$$\partial^2 p = c_2 \nabla_2 p \, \partial t^2$$

• p is the acoustic pressure.

• t is time.

• c is the speed of sound in the medium.

$y=a\sin(\omega t)$ is the equation for a simple harmonic motion of a transverse wave

16. Describe how the attack and decay phases of a sound envelope can influence the perception of rhythm and dynamics in music.
Attack: Time to reach maximum intensity.Decay: Reduction after the peak,
These distinguishes instruments from each other with the same pitch and volume.

17. How we can sound proof a room ?
Use dense materials, acoustic panels, carpets, and double-glazed windows.
Seal gaps and cracks.

18. Compare how sound propagates in air versus water.
Compression and rarefaction of molecules , speed of sound is higher in water than air, density and elasticity effect the sound transmission

19. How does frequency affect the perception of pitch, and what is the importance of this in speech communication?
Pitch: The perceived highness or lowness of sound.
human ear is more sensitive to mid frequency

20. What is pitch ? How frequency related with pitch ?
Pitch: The perceived highness or lowness of sound.
Higher frequency = higher pitch; lower frequency = lower pitch.

21. What is the function of eustachian tube ?
Balances pressure between the middle ear and the external environment.
Ensures proper vibration of the eardrum.

Part C

Essay Questions Answer any **2** Questions Each Question carries **10** marks

22. What are the characteristics of a sound wave . Explain with illustration ?

- Amplitude: Loudness (greater amplitude = louder sound).
- Frequency: Pitch (higher frequency = higher pitch).
- Wavelength: Distance between wave crests (related to pitch).
- Velocity: Speed of sound in the medium.

23. What are the factors that influence acoustic impedance, and how do these factors affect the efficiency of sound transmission?
Acoustical impedance is a measure of a material resistance to the propagation of sound waves

Density , stiffness, temperature, frequency, porosity these are the main factors affecting the acoustic impedance

impedance matching : to minimise reflection and maximise sound transmission, its often need to match the acoustic impedance of two materials

24. Describe the relationship between amplitude, loudness, and the decibel scale, and explain how they are used to measure and perceive sound.

- Amplitude: Loudness (greater amplitude = louder sound).
- Loudness, is a subjective perception of the intensity of a sound by the human ear
- Unit of loudness is phones ,
- The threshold of human hearing is OdB SPL, Threshold of pain120-140 dB SPL
- SPL meter is used to measure the sound pressure level
- Loudness of sound is measured in decibels (dB). This is actually a measure of intensity, which relates to how much energy the pressure wave has.